



NLP (Spring - 2026)

Assignment 3

Instructor

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Submission Guidelines:

- Submit your assignment on Google Classroom in the format "20XX.ipynb".
- The deadline is April **30 2026**, at 11:59 PM.

Declarations:

- Late submissions will incur penalties: 25% deduction on the first day, 50% on the second day, and zero marks thereafter.
- Plagiarism will result in zero marks for the assignment.
- This is a group assignment for a maximum of two students.
- Please ensure that you submit your own original work.

VIVA Policy:

- A VIVA (oral examination) will be conducted to assess your understanding of the assignment.
- The VIVA will be scheduled separately, and you will be notified of the date and time.
- Failure to attend the VIVA will result in zero marks for the assignment.

Academic Integrity:

- Plagiarism, collusion, and academic dishonesty will not be tolerated.
- Any instances of academic misconduct will be reported to the authorities and may result in severe penalties.

Objective

The objective of this assignment is to implement and compare four fine-tuning techniques for transformer-based models using a small-scale sentiment classification task. The assignment focuses on evaluating:

- Model accuracy
- Number of trainable parameters
- Training time
- GPU memory usage

This comparison will help identify trade-offs between Full Fine-Tuning and Parameter-Efficient Fine-Tuning (PEFT) methods, including LoRA, QLoRA and Adapter Tuning (IA³)

Models from Hugging Face

DistilBert

RoBERTa-Base

DeBERTa-v3

Dataset

- **Dataset:** IMDb Sentiment Classification
- **Source:** IMDb on Hugging Face (<https://huggingface.co/datasets/mteb/imdb>)

Tips

- Use **nvidia-smi** to measure GPU and memory usage
- Use Hugging Face **Trainer** API for consistency

Part 1 Data Preprocessing (10 marks)

1. Load the IMDb dataset using the Hugging Face **datasets** library.
2. Tokenize the dataset using the **model** tokenizer.
3. Pad and truncate sequences appropriately.

Part 2 Model Implementation (40 marks)

Implement and fine-tune the following models:

Method 1: Full Fine-Tuning (25% marks)

Method 2: LoRA Fine-Tuning (25% marks)

Method 3: QLoRA Fine-Tuning (25% marks)

Method 4: Adapter Tuning (IA³) (25% marks)

Insert IA³-style adapters and fine-tune them while keeping the rest of the model frozen.

Part 3. Ablation Studies (50 marks)

Evaluate how sensitive each fine-tuning method (**Full FT**, **LoRA**, **QLoRA**, **IA³**) is to key design/optimization choices under a **fixed compute budget**. Report both **performance** (Accuracy, F1-macro) and **efficiency** (training time, peak VRAM, trainable parameters) to identify **Pareto trade-offs**.

Experimental Controls

- **Compute budget:** same **max steps** and **total tokens** per run (use **gradient accumulation** to keep effective tokens constant).
- **Tokenizer policy:** `max_length = 512`, truncation/padding identical across runs.
- **Data:** IMDb sentiment (full) plus **low-resource** subset at **30%** (for best config).
- **Seeds:** **42, 43**; report **mean ± std**.
- **Metrics:** Accuracy, **F1-macro**, training time (s), **peak VRAM (GB)**, **trainable params (M)**.
- **Logging:** save CSV of configs + results and export plots (rank–accuracy, Pareto curves).

Ablation Factors

Each factor is varied **one-at-a-time**.

A. LoRA (encoders / seq2seq)

- **Rank (r):** 8, 16
- **Target modules:** `[q, v]` (compare to `[q, k, v, o]` on one run)
- **Placement:** Attention-only vs **Attention + FFN**
- **Alpha (α):** 16 (fixed unless exploring sensitivity)

B. QLoRA (*quantized base only*)

- **Quantization:** **NF4** (compare NF4 vs FP4 on one run)

- **Double quantization:** on (recommended)
- **LoRA rank on quantized base:** $r = 16$
- **Optimizer:** **Paged-AdamW** on (ablate off)

C. IA³ (adapters)

- **Location:** Attention-only vs **Attention + FFN**
- **Target tensors:** **values-only** (v-only)

D. Optimization (applies to all)

- **Learning rate (encoders):** **1e-5** → **2e-5** (single sensitivity probe)
- **Effective batch:** 16 (kept constant via grad accumulation)

Design Matrix (minimum set)

Global defaults: `max_length=256`, epochs=2, effective batch=16 (via grad accumulation), LR=1e-5 unless varied, data=100% unless varied, seeds = {42, 43}.

ExpID	Method	Factor	Levels / Config	Fixed Defaults	Purpose
01	Full FT	—	baseline	—	Accuracy/Time anchor
02	LoRA	Rank (r)	8	targets=[q,v], placement=Attn, $\alpha=16$	Low-rank PEFT point
03	LoRA	Rank (r)	16	targets=[q,v], placement=Attn, $\alpha=16$	Rank sensitivity
04	LoRA (2-way)	Placement	Attn → Attn+FFN (at $r=16$)	targets=[q,v], $\alpha=16$	Module placement effect

05	QLoRA	Quant/optim	NF4 , r=16, paged-AdamW=on , dq=on	—	Quantized PEFT efficiency
06	IA ³	Location	Attn → Attn+FFN	targets=v-only	Adapter placement effect
07	ALL	LR	1e-5 → 2e-5	method's best config so far	Optimization sensitivity
08 (<i>optional</i>)	ALL	Data fraction	100% → 30%	method's best config so far	Low-resource robustness

Per model (e.g., RoBERTa-Base and DistilBERT): run ExpID **01–07** with seeds {42, 43} and report mean ± std.

Include ExpID **08** (30% data) and/or a single LoRA **targets** breadth check [q,v] → [q,k,v,o].

Reporting (Example only)

Seed aggregation: mean ± std over seeds. Units: **Time (s)**, **VRAM (GB)**, **Params (M)**.

	Configuration (ExpID)	Accuracy (%)	F1 (macro)	Training Time (s)	Peak VRAM (GB)	Trainable Params (M)
01	Full FT (baseline)					
02	LoRA r=8, [q,v], Attn, α=16					
03	LoRA r=16, [q,v], Attn, α=16					

04	LoRA $r=16$, Attn+FFN , [q,v]					
05	QLoRA NF4 , $r=16$, dq=on, paged-AdamW=on					
06	IA³ Attn+FFN , v-only					
07	Best config with LR=2e-5					
08	Best config on 30% data					

Visualizations

- **LoRA rank** → **Accuracy/F1** (line plot; eg ExpID 02 vs 03).
- **Pareto curves**: (VRAM vs Accuracy) and (Time vs Accuracy) for all configurations.
- **Low-resource bar (optional)**: best-config @ 100% vs 30% data.

Submission Requirements

A. Notebook: 20XX.ipynb (80 marks)

- 1 Notebook with Proper implementation of all methods
- Results,metrics and visualizations included.

B. Final Comparative Report in IEEE Format: 20XX_Report.t.pdf (20 marks)

Sections:

1. Abstract
 - Brief overview of the task, methods used, and key findings
2. Introduction
 - Brief explanation of Full Fine-Tuning, LoRA, QLoRA, and IA³
3. Experimental Setup
 - Dataset description

- Hardware used
- Hyperparameters and configuration details
- 4. Results and Visualizations
 - Comparative charts and metrics
- 5. Analysis and Discussion (**what to discuss**)
 - Pareto frontier: Which configs dominate on accuracy vs VRAM/time?
 - LoRA rank sensitivity: Does $r = 16$ give meaningful gains over $r = 8$ relative to added params/time?
 - Module placement: Is Attention+FFN consistently better than Attention-only for LoRA/IA³?
 - QLoRA efficiency: How much VRAM and time does QLoRA save vs its accuracy delta?
 - Optimization: Does $LR=2e-5$ improve PEFT methods without destabilizing metrics?
 - Robustness (optional): Which method degrades least at 30% data?
- 6. Conclusion